AP Calculus AB Name: ______________________________

Unit 1, Lesson 4 - Continuity and One-Sided Limits

By the end of this lesson you should know how to:

- Determine continuity at a point and continuity on an open interval
- Determine one-sided limits and continuity on a closed interval
- Use properties of continuity
- Understand and use the Intermediate Value Theorem

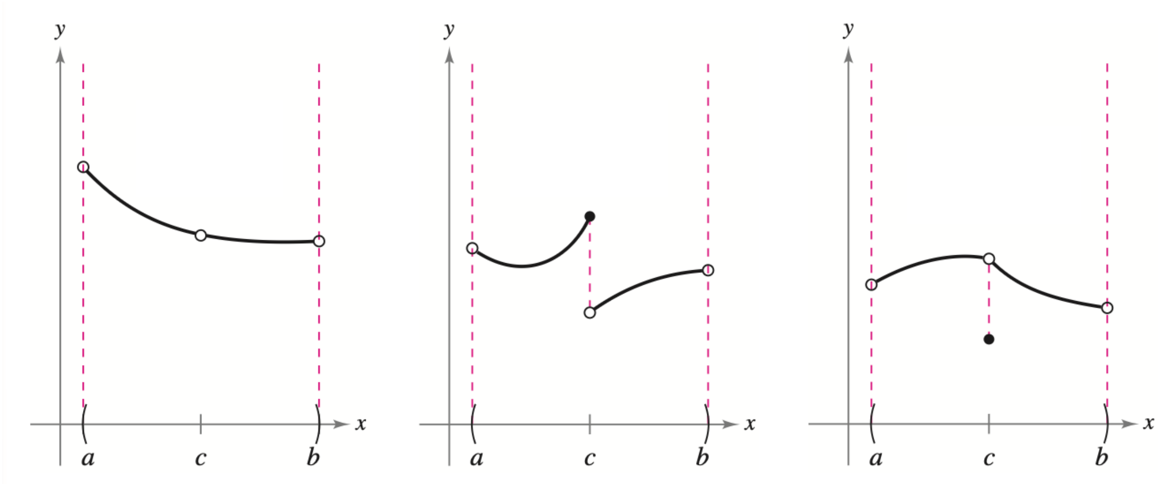
*In calculus, functions that behave predictably are often easier to analyze and work with. Often, this means that nearby inputs produce nearby outputs without sudden jumps, gaps, or interruptions in the graph. Functions with this type of local behavior are called **continuous**.*

Our goal for today is to understand what makes a function continuous at a point and on an interval, and to develop some important properties of continuous functions.

Before we venture into a proper definition of continuity, take a moment to visualize what you think "continuity" of a function might mean at a point and on an interval. Use the space below to create a few sketches of graphs with continuity.

After a minute, compare your drawings with your neighbors. What commonalities do you notice?

Let's look at some pictures that demonstrate what continuity is **not**.



What do you observe in each?

- In the first image, the function is not defined at $x = c$.
- In the second image, the limit does not exist at $x = c$
- In the third image, the function is defined and the limit at $x = c$ does exist, but these two quantities are not equal

If none of the above conditions are true, then the function f is called **continuous at c**, and this is formalized below in more affirmative language.

Definition of Continuity

Continuity at a Point

A function f is **continuous at c** when these three conditions are met.

1. $f(c)$ is defined.

2. $\lim_{x \to c} f(x)$ exists.

3. $\lim_{x \to c} f(x) = f(c)$

Continuity on an Open Interval

A function is **continuous on an open interval (a, b)** when the function is continuous at each point in the interval. A function that is continuous on the entire real number line $(-\infty, \infty)$ is **everywhere continuous.**

So, when a function is defined on an interval I (except possibly at c), and f is not continuous at c, then f is said to have a ***discontinuity*** at c. Discontinuities can be:

- ***removable*** if f can be made continuous by redefining $f(c)$, as is the case in the first and third examples above, or
- ***nonremovable***, as is the case in the second example above

Example 1. *Continuity of a Function*

Discuss the continuity of each function. Consider the domain as you make your decision.

a. $f(x) = \frac{1}{x}$

b. $g(x) = \frac{x^2-1}{x-1}$

c. $h(x) = \begin{cases} x+1, & x \leq 0 \\ x^2+1, & x > 0 \end{cases}$

d. $y = \sin x$

One-Sided Limits and Continuity on a Closed Interval

To understand continuity on a closed interval, we'll need to expand our knowledge of limits to include something called a ***one-sided limit***. This is where we will carefully consider the behavior of a function as x approaches c from either the right-hand or left-hand side.

We'll use a "superscript" notation to indicate whether x is approaching c from the right (positive) side or the left (negative) side.

$\lim_{x \to c^+} f(x)$ "the limit of $f(x)$ as x approaches c from the right"

$\lim_{x \to c^-} f(x)$ "the limit of $f(x)$ as x approaches c from the left"

Such one-sided limits are especially helpful in taking limits of:

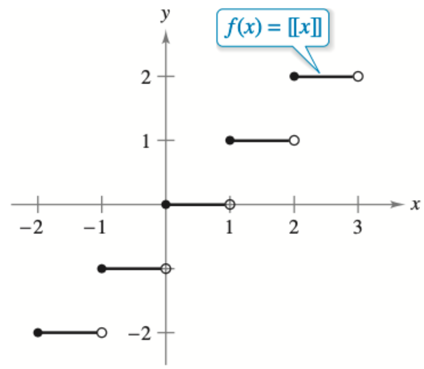
- radical functions of even index (since their domains are limited)
- piecewise functions at "splitting points"
- vertical asymptotes, whose behavior may vary on either side of the asymptote

Example 2. *A One-Sided Limit*

Consider the function $f(x) = \sqrt{4 - x^2}$. Calculate $\lim_{x \to 2^-} f(x)$ and $\lim_{x \to 2^+} f(x)$. Explain any differences in your answers.

Example 3. *The Greatest Integer Function*

Find the limit of the greatest integer function $f(x) = [\![x]\!]$ as x approaches 0 from the left and from the right. Please use proper notation.



Greatest integer function

What would you say about $\lim_{x \to 0} [\![x]\!]$?

When the limit from the left is not equal to the limit from the right, the (two-sided) limit does not exist, as stated formally below.

THEOREM 1.10 The Existence of a Limit

Let f be a function, and let c and L be real numbers. The limit of $f(x)$ as x approaches c is L if and only if

$$\lim_{x \to c^-} f(x) = L \quad \text{and} \quad \lim_{x \to c^+} f(x) = L.$$

What about continuity on a ***closed interval***? We can claim that a function is continuous on a closed interval when it is continuous in the interior of the interval and exhibits one-sided continuity at the endpoints, as stated formally below.

Definition of Continuity on a Closed Interval

A function f is **continuous on the closed interval** $[a, b]$ when f is continuous on the open interval (a, b) and

$$\lim_{x \to a^+} f(x) = f(a)$$

and

$$\lim_{x \to b^-} f(x) = f(b).$$

The function f is **continuous from the right** at a and **continuous from the left** at b (see Figure 1.31).

We can create similar definitions to cover functions on intervals that are neither open nor closed. For example, $f(x) = \sqrt{x}$ is continuous on the infinite interval $[0, \infty)$, while $g(x) = \sqrt{2-x}$ is continuous on the infinite interval $(-2, \infty]$.

Example 4. *Continuity on a Closed Interval*

Discuss the continuity of $f(x) = \sqrt{1-x^2}$.

Once you feel comfortable working with continuity, you'll soon discover that there are certain properties of continuity that allow us to combine functions in a way that preserves continuity. In particular, if two functions, f and g, are continuous (which are continuous at $x = c$), and b is a real number, then the following functions are also continuous at c.

1. Scalar multiple: bf
2. Sum or difference: $f \pm g$
3. Product: fg
4. Quotient: $\frac{f}{g}$, provided $g(c) \neq 0$

Additionally, the following functions are continuous at every point in their domains:

1. Polynomial: $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$
2. Rational: $r(x) = \frac{p(x)}{q(x)}$, provided $q(x) \neq 0$
3. Radical: $f(x) = \sqrt[n]{x}$
4. Trigonometric: $\sin x, \cos x, \tan x, \cot x, \sec x, \csc x$

Example 6. *Applying Properties of Continuity*

By the previous statement, we can conclude that each of the following functions is continuous at every point in its domain.

$$f(x) = x + \sin x \qquad f(x) = 3\tan x \qquad f(x) = \frac{x^2+1}{\cos x}$$

Additionally, we can make some observation about the continuity of composite functions.

If g is continuous at c and f is continuous at $f(c)$, then the *composite* function given by $f(g(x))$ is continuous at c.

This allows us to determine intervals of continuity for such functions as $f(x) = \sin 3x$, $f(x) = \sqrt{x^2+1}$, and $f(x) = \tan \frac{1}{x}$.

Example 7. *Testing for Continuity*

Describe the interval(s) on which each function is continuous.

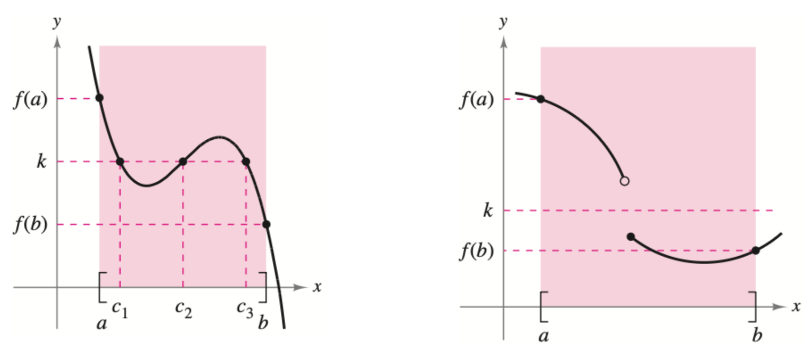
a. $f(x) = \tan x$ b. $g(x) = \begin{cases} \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ c. $h(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

The Intermediate Value Theorem

The Intermediate Value Theorem is a theorem that says "if f is a continuous function on $[a, b]$, then every value between $f(a)$ and $f(b)$ is represented on $[a, b]$". When it's phrased more formally, it guarantees a specific value without providing a method of finding it. Such theorems are known as ***existence theorems***.

Here's the formal version:

If f is continuous on the closed interval $[a, b]$, $f(a) \neq f(b)$, and k is any number between $f(a)$ and $f(b)$, then there is at least one real number c in $[a, b]$ such that $f(c) = k$.



f is continuous on $[a, b]$.
[There exist three c's such that $f(c) = k$.]

f is not continuous on $[a, b]$.
[There are no c's such that $f(c) = k$.]

Example 8. *An Application of the Intermediate Value Theorem*

Use the Intermediate Value Theorem to show that the polynomial function $f(x) = x^3 + 2x - 1$ has a zero in the interval $[0,1]$.

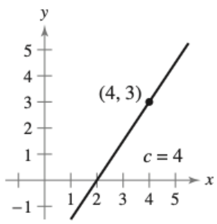
Note: f is a continuous function on $[0,1]$, so we can apply the Intermediate Value Theorem.

1.4 Exercises

See CalcChat.com for tutorial help and worked-out solutions to odd-numbered exercises.

Limits and Continuity In Exercises 1–6, use the graph to determine the limit, and discuss the continuity of the function.

(a) $\lim\limits_{x\to c^+} f(x)$ (b) $\lim\limits_{x\to c^-} f(x)$ (c) $\lim\limits_{x\to c} f(x)$

1.


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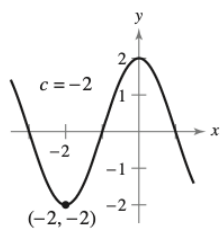

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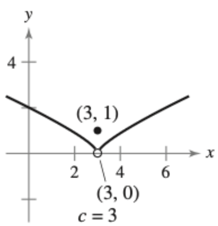

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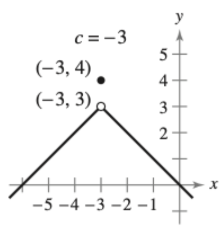

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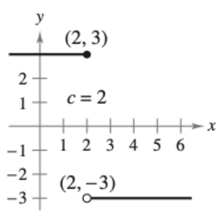

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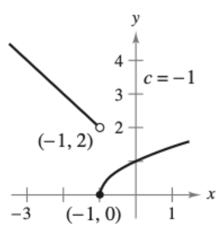


Finding a Limit In Exercises 7–26, find the limit (if it exists). If it does not exist, explain why.

7. $\lim\limits_{x\to 8^+} \dfrac{1}{x+8}$

8. $\lim\limits_{x\to 2^-} \dfrac{2}{x+2}$

9. $\lim\limits_{x\to 5^+} \dfrac{x-5}{x^2-25}$

10. $\lim\limits_{x\to 4^+} \dfrac{4-x}{x^2-16}$

11. $\lim\limits_{x\to -3^-} \dfrac{x}{\sqrt{x^2-9}}$

12. $\lim\limits_{x\to 4^-} \dfrac{\sqrt{x}-2}{x-4}$

13. $\lim\limits_{x\to 0^-} \dfrac{|x|}{x}$

14. $\lim\limits_{x\to 10^+} \dfrac{|x-10|}{x-10}$

15. $\lim\limits_{\Delta x\to 0^-} \dfrac{\dfrac{1}{x+\Delta x}-\dfrac{1}{x}}{\Delta x}$

16. $\lim\limits_{\Delta x\to 0^+} \dfrac{(x+\Delta x)^2+x+\Delta x-(x^2+x)}{\Delta x}$

17. $\lim\limits_{x\to 3^-} f(x)$, where $f(x) = \begin{cases} \dfrac{x+2}{2}, & x \le 3 \\ \dfrac{12-2x}{3}, & x > 3 \end{cases}$

18. $\lim\limits_{x\to 3} f(x)$, where $f(x) = \begin{cases} x^2-4x+6, & x < 3 \\ -x^2+4x-2, & x \ge 3 \end{cases}$

19. $\lim\limits_{x\to 1} f(x)$, where $f(x) = \begin{cases} x^3+1, & x < 1 \\ x+1, & x \ge 1 \end{cases}$

20. $\lim\limits_{x\to 1^+} f(x)$, where $f(x) = \begin{cases} x, & x \le 1 \\ 1-x, & x > 1 \end{cases}$

21. $\lim\limits_{x\to \pi} \cot x$

22. $\lim\limits_{x\to \pi/2} \sec x$

23. $\lim\limits_{x\to 4^-} (5[\![x]\!]-7)$

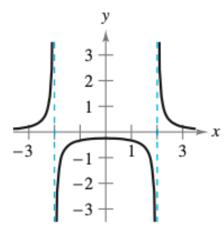
24. $\lim\limits_{x\to 2^+} (2x-[\![x]\!])$

25. $\lim\limits_{x\to 3} (2-[\![-x]\!])$

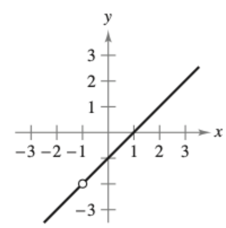
26. $\lim\limits_{x\to 1} \left(1-\left[\!\!\left[-\dfrac{x}{2}\right]\!\!\right]\right)$

Continuity of a Function In Exercises 27–30, discuss the continuity of each function.

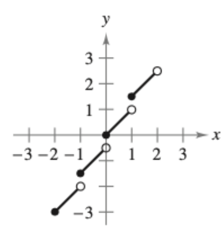
27. $f(x) = \dfrac{1}{x^2-4}$



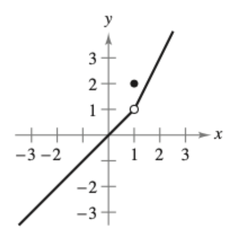
28. $f(x) = \dfrac{x^2-1}{x+1}$



29. $f(x) = \frac{1}{2}[\![x]\!] + x$



30. $f(x) = \begin{cases} x, & x < 1 \\ 2, & x = 1 \\ 2x-1, & x > 1 \end{cases}$



Continuity on a Closed Interval In Exercises 31–34, discuss the continuity of the function on the closed interval.

Function	Interval
31. $g(x) = \sqrt{49-x^2}$	$[-7, 7]$
32. $f(t) = 3-\sqrt{9-t^2}$	$[-3, 3]$
33. $f(x) = \begin{cases} 3-x, & x \le 0 \\ 3+\frac{1}{2}x, & x > 0 \end{cases}$	$[-1, 4]$
34. $g(x) = \dfrac{1}{x^2-4}$	$[-1, 2]$

Removable and Nonremovable Discontinuities In Exercises 35–60, find the x-values (if any) at which f is not continuous. Which of the discontinuities are removable?

35. $f(x) = \dfrac{6}{x}$

36. $f(x) = \dfrac{4}{x-6}$

37. $f(x) = x^2-9$

38. $f(x) = x^2-4x+4$

39. $f(x) = \dfrac{1}{4 - x^2}$

40. $f(x) = \dfrac{1}{x^2 + 1}$

41. $f(x) = 3x - \cos x$

42. $f(x) = \cos \dfrac{\pi x}{2}$

43. $f(x) = \dfrac{x}{x^2 - x}$

44. $f(x) = \dfrac{x}{x^2 - 4}$

45. $f(x) = \dfrac{x}{x^2 + 1}$

46. $f(x) = \dfrac{x - 5}{x^2 - 25}$

47. $f(x) = \dfrac{x + 2}{x^2 - 3x - 10}$

48. $f(x) = \dfrac{x + 2}{x^2 - x - 6}$

49. $f(x) = \dfrac{|x + 7|}{x + 7}$

50. $f(x) = \dfrac{|x - 5|}{x - 5}$

51. $f(x) = \begin{cases} x, & x \le 1 \\ x^2, & x > 1 \end{cases}$

52. $f(x) = \begin{cases} -2x + 3, & x < 1 \\ x^2, & x \ge 1 \end{cases}$

53. $f(x) = \begin{cases} \frac{1}{2}x + 1, & x \le 2 \\ 3 - x, & x > 2 \end{cases}$

54. $f(x) = \begin{cases} -2x, & x \le 2 \\ x^2 - 4x + 1, & x > 2 \end{cases}$

55. $f(x) = \begin{cases} \tan \dfrac{\pi x}{4}, & |x| < 1 \\ x, & |x| \ge 1 \end{cases}$

56. $f(x) = \begin{cases} \csc \dfrac{\pi x}{6}, & |x - 3| \le 2 \\ 2, & |x - 3| > 2 \end{cases}$

57. $f(x) = \csc 2x$

58. $f(x) = \tan \dfrac{\pi x}{2}$

59. $f(x) = [\![x - 8]\!]$

60. $f(x) = 5 - [\![x]\!]$

Making a Function Continuous **In Exercises 61–66, find the constant a, or the constants a and b, such that the function is continuous on the entire real number line.**

61. $f(x) = \begin{cases} 3x^2, & x \ge 1 \\ ax - 4, & x < 1 \end{cases}$

62. $f(x) = \begin{cases} 3x^3, & x \le 1 \\ ax + 5, & x > 1 \end{cases}$

63. $f(x) = \begin{cases} x^3, & x \le 2 \\ ax^2, & x > 2 \end{cases}$

64. $g(x) = \begin{cases} \dfrac{4 \sin x}{x}, & x < 0 \\ a - 2x, & x \ge 0 \end{cases}$

65. $f(x) = \begin{cases} 2, & x \le -1 \\ ax + b, & -1 < x < 3 \\ -2, & x \ge 3 \end{cases}$

66. $g(x) = \begin{cases} \dfrac{x^2 - a^2}{x - a}, & x \ne a \\ 8, & x = a \end{cases}$

Continuity of a Composite Function **In Exercises 67–72, discuss the continuity of the composite function $h(x) = f(g(x))$.**

67. $f(x) = x^2$
$g(x) = x - 1$

68. $f(x) = 5x + 1$
$g(x) = x^3$

69. $f(x) = \dfrac{1}{x - 6}$
$g(x) = x^2 + 5$

70. $f(x) = \dfrac{1}{\sqrt{x}}$
$g(x) = x - 1$

71. $f(x) = \tan x$
$g(x) = \dfrac{x}{2}$

72. $f(x) = \sin x$
$g(x) = x^2$

Finding Discontinuities **In Exercises 73–76, use a graphing utility to graph the function. Use the graph to determine any x-values at which the function is not continuous.**

73. $f(x) = [\![x]\!] - x$

74. $h(x) = \dfrac{1}{x^2 + 2x - 15}$

75. $g(x) = \begin{cases} x^2 - 3x, & x > 4 \\ 2x - 5, & x \le 4 \end{cases}$

76. $f(x) = \begin{cases} \dfrac{\cos x - 1}{x}, & x < 0 \\ 5x, & x \ge 0 \end{cases}$

Testing for Continuity **In Exercises 77–84, describe the interval(s) on which the function is continuous.**

77. $f(x) = \dfrac{x}{x^2 + x + 2}$

78. $f(x) = \dfrac{x + 1}{\sqrt{x}}$

79. $f(x) = 3 - \sqrt{x}$

80. $f(x) = x\sqrt{x + 3}$

81. $f(x) = \sec \dfrac{\pi x}{4}$

82. $f(x) = \cos \dfrac{1}{x}$

83. $f(x) = \begin{cases} \dfrac{x^2 - 1}{x - 1}, & x \ne 1 \\ 2, & x = 1 \end{cases}$

84. $f(x) = \begin{cases} 2x - 4, & x \ne 3 \\ 1, & x = 3 \end{cases}$

Writing **In Exercises 85 and 86, use a graphing utility to graph the function on the interval $[-4, 4]$. Does the graph of the function appear to be continuous on this interval? Is the function continuous on $[-4, 4]$? Write a short paragraph about the importance of examining a function analytically as well as graphically.**

85. $f(x) = \dfrac{\sin x}{x}$

86. $f(x) = \dfrac{x^3 - 8}{x - 2}$

Writing **In Exercises 87–90, explain why the function has a zero in the given interval.**

Function	Interval
87. $f(x) = \frac{1}{12}x^4 - x^3 + 4$	$[1, 2]$
88. $f(x) = x^3 + 5x - 3$	$[0, 1]$
89. $f(x) = x^2 - 2 - \cos x$	$[0, \pi]$
90. $f(x) = -\dfrac{5}{x} + \tan\left(\dfrac{\pi x}{10}\right)$	$[1, 4]$

Using the Intermediate Value Theorem **In Exercises 91–94, use the Intermediate Value Theorem and a graphing utility to approximate the zero of the function in the interval $[0, 1]$. Repeatedly "zoom in" on the graph of the function to approximate the zero accurate to two decimal places. Use the *zero* or *root* feature of the graphing utility to approximate the zero accurate to four decimal places.**

91. $f(x) = x^3 + x - 1$

92. $f(x) = x^4 - x^2 + 3x - 1$

93. $g(t) = 2\cos t - 3t$

94. $h(\theta) = \tan\theta + 3\theta - 4$

Using the Intermediate Value Theorem **In Exercises 95–98, verify that the Intermediate Value Theorem applies to the indicated interval and find the value of c guaranteed by the theorem.**

95. $f(x) = x^2 + x - 1, \quad [0, 5], \quad f(c) = 11$

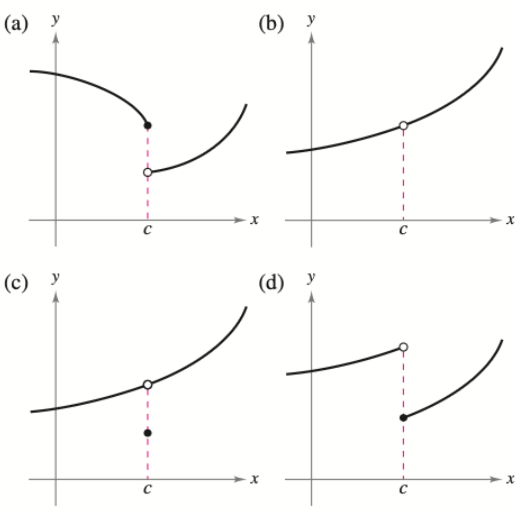
96. $f(x) = x^2 - 6x + 8, \quad [0, 3], \quad f(c) = 0$

97. $f(x) = x^3 - x^2 + x - 2, \quad [0, 3], \quad f(c) = 4$

98. $f(x) = \dfrac{x^2 + x}{x - 1}, \quad \left[\dfrac{5}{2}, 4\right], \quad f(c) = 6$

WRITING ABOUT CONCEPTS

99. Using the Definition of Continuity State how continuity is destroyed at $x = c$ for each of the following graphs.



100. Sketching a Graph Sketch the graph of any function f such that

$$\lim_{x\to 3^+} f(x) = 1 \quad \text{and} \quad \lim_{x\to 3^-} f(x) = 0.$$

Is the function continuous at $x = 3$? Explain.

101. Continuity of Combinations of Functions If the functions f and g are continuous for all real x, is $f + g$ always continuous for all real x? Is f/g always continuous for all real x? If either is not continuous, give an example to verify your conclusion.

102. Removable and Nonremovable Discontinuities Describe the difference between a discontinuity that is removable and one that is nonremovable. In your explanation, give examples of the following descriptions.

(a) A function with a nonremovable discontinuity at $x = 4$

(b) A function with a removable discontinuity at $x = -4$

(c) A function that has both of the characteristics described in parts (a) and (b)

True or False? **In Exercises 103–106, determine whether the statement is true or false. If it is false, explain why or give an example that shows it is false.**

103. If $\lim_{x\to c} f(x) = L$ and $f(c) = L$, then f is continuous at c.

104. If $f(x) = g(x)$ for $x \neq c$ and $f(c) \neq g(c)$, then either f or g is not continuous at c.

105. A rational function can have infinitely many x-values at which it is not continuous.

106. The function

$$f(x) = \frac{|x - 1|}{x - 1}$$

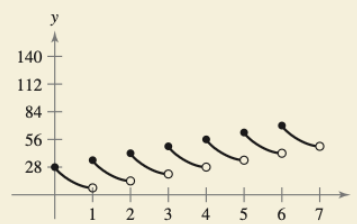
is continuous on $(-\infty, \infty)$.

107. Think About It Describe how the functions

$$f(x) = 3 + [\![x]\!] \quad \text{and} \quad g(x) = 3 - [\![-x]\!]$$

differ.

108. HOW DO YOU SEE IT? Every day you dissolve 28 ounces of chlorine in a swimming pool. The graph shows the amount of chlorine $f(t)$ in the pool after t days. Estimate and interpret $\lim_{t\to 4^-} f(t)$ and $\lim_{t\to 4^+} f(t)$.



109. Telephone Charges A long distance phone service charges \$0.40 for the first 10 minutes and \$0.05 for each additional minute or fraction thereof. Use the greatest integer function to write the cost C of a call in terms of time t (in minutes). Sketch the graph of this function and discuss its continuity.

110. Inventory Management

The number of units in inventory in a small company is given by

$$N(t) = 25\left(2\left[\!\left[\frac{t+2}{2}\right]\!\right] - t\right)$$

where t is the time in months. Sketch the graph of this function and discuss its continuity. How often must this company replenish its inventory?

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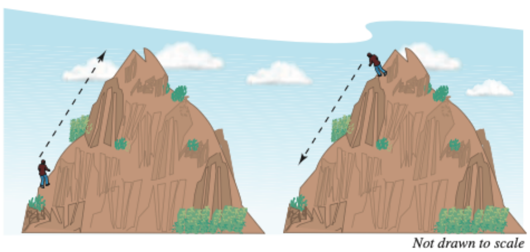
111. Déjà Vu At 8:00 A.M. on Saturday, a man begins running up the side of a mountain to his weekend campsite (see figure). On Sunday morning at 8:00 A.M., he runs back down the mountain. It takes him 20 minutes to run up, but only 10 minutes to run down. At some point on the way down, he realizes that he passed the same place at exactly the same time on Saturday. Prove that he is correct. [*Hint:* Let $s(t)$ and $r(t)$ be the position functions for the runs up and down, and apply the Intermediate Value Theorem to the function $f(t) = s(t) - r(t)$.]



112. Volume Use the Intermediate Value Theorem to show that for all spheres with radii in the interval $[5, 8]$, there is one with a volume of 1500 cubic centimeters.

113. Proof Prove that if f is continuous and has no zeros on $[a, b]$, then either

$$f(x) > 0 \text{ for all } x \text{ in } [a, b] \quad \text{or} \quad f(x) < 0 \text{ for all } x \text{ in } [a, b].$$

114. Dirichlet Function Show that the Dirichlet function

$$f(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ 1, & \text{if } x \text{ is irrational} \end{cases}$$

is not continuous at any real number.

115. Continuity of a Function Show that the function

$$f(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ kx, & \text{if } x \text{ is irrational} \end{cases}$$

is continuous only at $x = 0$. (Assume that k is any nonzero real number.)

116. Signum Function The **signum function** is defined by

$$\operatorname{sgn}(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0. \\ 1, & x > 0 \end{cases}$$

Sketch a graph of $\operatorname{sgn}(x)$ and find the following (if possible).

(a) $\lim_{x \to 0^-} \operatorname{sgn}(x)$ (b) $\lim_{x \to 0^+} \operatorname{sgn}(x)$ (c) $\lim_{x \to 0} \operatorname{sgn}(x)$

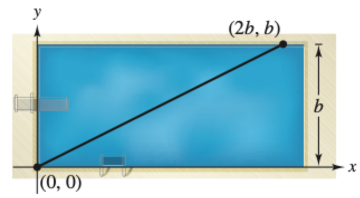
117. Modeling Data The table lists the speeds S (in feet per second) of a falling object at various times t (in seconds).

t	0	5	10	15	20	25	30
S	0	48.2	53.5	55.2	55.9	56.2	56.3

(a) Create a line graph of the data.

(b) Does there appear to be a limiting speed of the object? If there is a limiting speed, identify a possible cause.

118. Creating Models A swimmer crosses a pool of width b by swimming in a straight line from $(0, 0)$ to $(2b, b)$. (See figure.)



(a) Let f be a function defined as the y-coordinate of the point on the long side of the pool that is nearest the swimmer at any given time during the swimmer's crossing of the pool. Determine the function f and sketch its graph. Is f continuous? Explain.

(b) Let g be the minimum distance between the swimmer and the long sides of the pool. Determine the function g and sketch its graph. Is g continuous? Explain.

119. Making a Function Continuous Find all values of c such that f is continuous on $(-\infty, \infty)$.

$$f(x) = \begin{cases} 1 - x^2, & x \le c \\ x, & x > c \end{cases}$$

120. Proof Prove that for any real number y there exists x in $(-\pi/2, \pi/2)$ such that $\tan x = y$.

121. Making a Function Continuous Let

$$f(x) = \frac{\sqrt{x + c^2} - c}{x}, \quad c > 0.$$

What is the domain of f? How can you define f at $x = 0$ in order for f to be continuous there?

122. Proof Prove that if

$$\lim_{\Delta x \to 0} f(c + \Delta x) = f(c)$$

then f is continuous at c.

123. Continuity of a Function Discuss the continuity of the function $h(x) = x[\![x]\!]$.

124. Proof

(a) Let $f_1(x)$ and $f_2(x)$ be continuous on the closed interval $[a, b]$. If $f_1(a) < f_2(a)$ and $f_1(b) > f_2(b)$, prove that there exists c between a and b such that $f_1(c) = f_2(c)$.

(b) Show that there exists c in $\left[0, \frac{\pi}{2}\right]$ such that $\cos x = x$. Use a graphing utility to approximate c to three decimal places.

PUTNAM EXAM CHALLENGE

125. Prove or disprove: If x and y are real numbers with $y \ge 0$ and $y(y + 1) \le (x + 1)^2$, then $y(y - 1) \le x^2$.

126. Determine all polynomials $P(x)$ such that

$$P(x^2 + 1) = (P(x))^2 + 1 \text{ and } P(0) = 0.$$

These problems were composed by the Committee on the Putnam Prize Competition.
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